

Assessment 2

Network Troubleshooting

Section A: Scenario-Based Concept Application

1. Explain how Deep Packet Inspection (DPI) at the Network and Transport layers of the OSI model allows a startup to isolate specific traffic bottlenecks causing slow internet speeds despite high available bandwidth.

- Deep Packet Inspection (DPI) examines packet headers and payloads at the **Network Layer (Layer 3)** and **Transport Layer (Layer 4)** of the OSI model.
 - At **Layer 3**, DPI analyzes:
 - Source and destination IP addresses
 - Routing paths
 - Network congestion
 - At **Layer 4**, DPI checks:
 - TCP/UDP port numbers
 - Session information
 - Application-specific traffic patterns
- In a startup office, internet speed may appear slow even when bandwidth usage is low. DPI helps identify:
 - Excessive video streaming traffic
 - Malware communication
 - Large downloads
 - Congested TCP sessions
 - High retransmissions

2. Explain the distinct roles of DHCP and DNS in a scenario where startup devices are successfully assigned IP addresses but remain unable to resolve external web addresses for internet access.

❖ **DHCP (Dynamic Host Configuration Protocol)**

- DHCP automatically provides:

- IP address
- Subnet mask
- Default gateway
- DNS server address

❖ **DNS (Domain Name System)**

- DNS converts domain names into IP addresses.

- A device receives:
 - IP: 192.168.10.20
 - Gateway: 192.168.10.1
- But when opening a website, it fails.

❖ Reason:

- DHCP is working correctly because the device received an IP.
- DNS is failing because domain names like google.com cannot be translated into IP addresses.

3. Explain the necessity of Inter-VLAN Routing and the specific role of static routes when a startup requires communication between two isolated departments on the same physical switch.

- VLANs isolate departments into separate broadcast domains.

Example:

- VLAN 10 → Marketing
- VLAN 20 → Engineering

- Even on the same physical switch, devices in different VLANs cannot communicate directly.

❖ **Inter-VLAN Routing**

- A router or Layer 3 switch is required to route traffic between VLANs.

Purpose:

- Enables communication between isolated departments
- Maintains security and traffic separation

❖ **Role of Static Routes**

- Static routes define the exact path traffic should follow.

Example:

- VLAN 10 subnet: 192.168.10.0/27
- VLAN 20 subnet: 192.168.10.32/27

4. Explain the performance differences between ICMP and HTTP/HTTPS protocols that would cause a website to experience high latency while a ping test returns a fast, successful response.

❖ **ICMP (Ping)**

- Ping uses ICMP packets:

- Very small size
- Minimal processing
- No webpage content

❖ **HTTP/HTTPS**

- Web traffic involves:

- TCP handshake
- SSL/TLS encryption
- Large webpage downloads
- Server-side processing
- Images, scripts, and advertisements

- ❖ websites may still load slowly because:
 - DNS resolution delay
 - Web server overload
 - TCP retransmissions
 - HTTPS encryption overhead
 - Congested application traffic
- Therefore:
 - Fast ping only proves basic connectivity.
 - It does not guarantee web application performance.

5. Explain the conceptual logic of Firewall Access Control Lists (ACLs) and the strategic placement of a firewall to restrict unauthorized traffic between a startup's management and general staff departments.

❖ **Firewall ACL Logic**

- Access Control Lists (ACLs) are rules that:
 - Permit specific traffic
 - Deny unauthorized traffic

❖ ACLs evaluate:

- Source IP
- Destination IP
- Protocol
- Port number

❖ Example:

- Management VLAN: 192.168.10.0/27
- Staff VLAN: 192.168.20.0/27

❖ ACL Rule:

- Allow Management → Staff
- Deny Staff → Management

❖ Benefits:

- Prevents unauthorized access
- Limits malware spread
- Protects sensitive company systems

Section B: Practical Application

1. Design a VLSM subnet scheme for 3 departments requiring 15 hosts each using \$192.168.10.0/24

- Need at least 15 hosts:
- $2^5 - 2 = 30$
- ❖ Therefore:
- 5 host bits required
- Subnet mask = /27

❖ Subnet Mask table :

Department	Network Address	Subnet Mask	Host Range	Broadcast
VLAN 10	192.168.10.0/27	255.255.255.224	192.168.10.1 – 30	192.168.10.31
VLAN 20	192.168.10.32/27	255.255.255.224	192.168.10.33 – 62	192.168.10.63
VLAN 30	192.168.10.64/27	255.255.255.224	192.168.10.65 – 94	192.168.10.95

2. Configure VLANs 10, 20, and 30 on a Cisco switch and assign physical interfaces to each.

```
enable
configure terminal
```

```
vlan 10
name Marketing
```

```
vlan 20
name Engineering
```

```
vlan 30
name HR
exit
```

```
interface fastethernet0/1
switchport mode access
switchport access vlan 10
```

```
interface fastethernet0/2
switchport mode access
switchport access vlan 20
interface fastethernet0/3
switchport mode access
switchport access vlan 30
```

3. Set up a DHCP pool on a Router/L3 switch to provide dynamic addressing for the VLAN 10 subnet.

```
ip dhcp pool vlan10
network 192.168.10.0 255.255.255.224
default-router 192.168.10.1
dns-server 8.8.8.8
do wr
ex
do wr
```

4. Capture a network trace in Wireshark and identify the IP addresses in the DNS query and HTTP GET headers.

- ❖ DNS Query
- ❖ Look for:
 - Source IP = PC address
 - Destination IP = DNS server
 - Query Name = website domain
- ❖ Example:
 - Source: 192.168.10.10
 - Destination: 8.8.8.8
- ❖ HTTP GET Request
- ❖ Look for:
 - HTTP GET header
 - Source and destination IPs
- ❖ Example:
 - Source: 192.168.10.10
 - Destination: 142.250.183.14

Section C: Integrated Case Study

Scenario: TechLaunch, a startup with 40 employees, is experiencing "flapping" network connectivity. Employees in the Marketing wing report that their internet drops for 30 seconds every few minutes. The Engineering wing, located on a different VLAN, remains perfectly stable. A ping test from a Marketing PC to the Core Switch shows a consistent 25% packet loss and fluctuating latency. Questions:

1. Based on the symptoms, identify the two most likely OSI layers where this fault is occurring.

✚ The problem most likely exists at:

❖ **Layer 1 — Physical Layer**

- Possible causes:
 - Bad cable
 - Loose connector
 - Duplex mismatch
 - Faulty switch port

❖ **Layer 2 — Data Link Layer**

- Possible causes:
 - Switching loop
 - MAC table instability
 - Broadcast storm
 - VLAN issue

2. Outline a 4-step "Bottom-Up" troubleshooting sequence to isolate this specific issue.

Step 1 Check physical connections:

- Cables
- LEDs
- NIC status

Step 2 Check switch port statistics:

- Errors
- Collisions
- CRC failures

Step 3 Verify VLAN configuration:

- Correct VLAN assignment
- Trunk status

Step 4 Analyze traffic:

- Use Wireshark

- Look for broadcast storms or duplicate frames
3. List three CLI commands you would run on the switch to check for cable faults or broadcast storms.

❖ Check interface errors:

- o `show interfaces`

❖ Check MAC address instability:

- o `show mac address-table`

❖ Check spanning-tree status:

- o `show spanning-tree`

❖ Additional useful command:

- o `show interfaces counters errors`

4. If the issue is a "Switching Loop," explain which protocol should have prevented this and why it might have failed.

➤ The protocol that should prevent switching loops is:

❖ **Spanning Tree Protocol (STP)**

- Purpose:
 - Blocks redundant paths
 - Prevents broadcast storms
 - Prevents MAC table instability
- Why it may fail:
 - STP disabled
 - Misconfigured priorities
 - Faulty trunk links
 - Rapid topology changes

5. Propose a long-term configuration change to ensure an error in one department doesn't impact the rest of the office.

❖ Implement:

- Proper VLAN segmentation
- STP protections
- Port security
- Storm control
- Layer 3 routing boundaries

❖ Benefits:

- Problems remain isolated within one department

- Broadcast storms cannot spread network-wide
- Improved startup network stability and scalability